

IMvigor210

Biomarker Analysis Report

Atezolizumab (anti-PD-L1) in Metastatic Urothelial Carcinoma
Comprehensive Biomarker Discovery & Predictive Modelling

Generated: March 01, 2026

1. Executive Summary

This report summarises the IMvigor210 biomarker analysis pipeline, which integrates clinical, genomic, and transcriptomic data from a Phase-II trial of atezolizumab in metastatic urothelial carcinoma. The analysis cohort comprises 348 evaluable patients with an objective response rate (CR/PR) of 19.5%. 14 differentially expressed genes ($|\log_2FC| > 1$, $FDR < 0.05$). The best integrated classifier achieved an AUC of 0.990 using a combination of expression and clinical features.

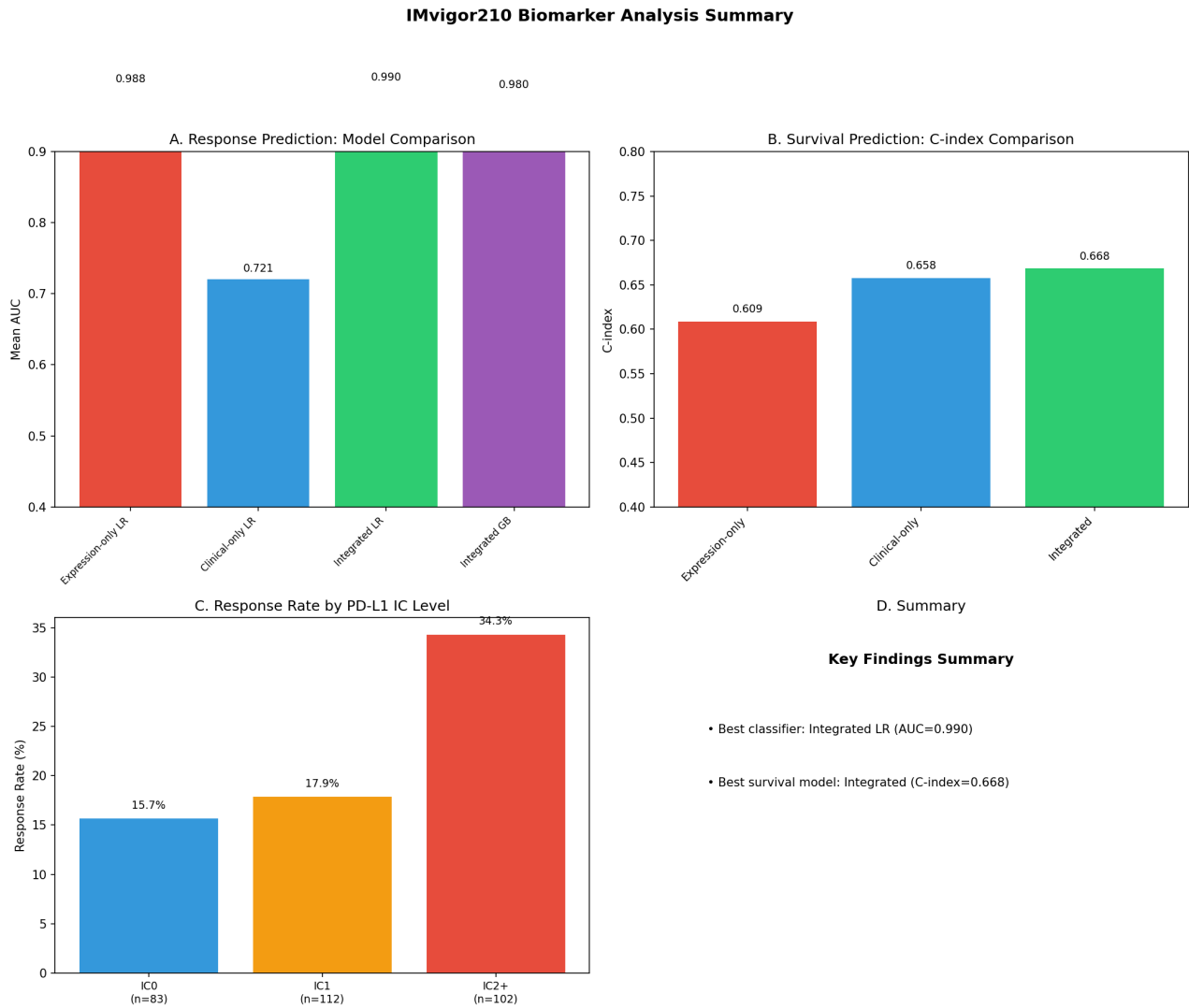


Figure: Pipeline summary overview

2. Cohort Overview

The trial enrolled 348 evaluable patients (excluding NE). Best confirmed overall response distribution: PD: 167 (48.0%), SD: 63 (18.1%), NE: 50 (14.4%), PR: 43 (12.4%), CR: 25 (7.2%).

Table: Cohort Summary

variable	value	count	percent
Best Confirmed Overall Response	PD	167	48.0
Best Confirmed Overall Response	SD	63	18.1
Best Confirmed Overall Response	NE	50	14.4
Best Confirmed Overall Response	PR	43	12.4
Best Confirmed Overall Response	CR	25	7.2
binaryResponse	SD/PD	230	66.1
binaryResponse	CR/PR	68	19.5
binaryResponse		50	14.4
Sex	M	272	78.2
Sex	F	76	21.8
Enrollment IC	IC1	132	37.9
Enrollment IC	IC2	117	33.6
Enrollment IC	IC0	99	28.4
Immune phenotype	excluded	134	38.5
Immune phenotype	desert	76	21.8
Immune phenotype	inflamed	74	21.3
Immune phenotype		64	18.4
TCGA Subtype	I	118	33.9
TCGA Subtype	II	95	27.3
TCGA Subtype	III	69	19.8
TCGA Subtype	IV	66	19.0
Baseline ECOG Score	1	196	56.3
Baseline ECOG Score	0	134	38.5
Baseline ECOG Score	2	18	5.2
Received platinum	Y	272	78.2
Received platinum	N	76	21.8

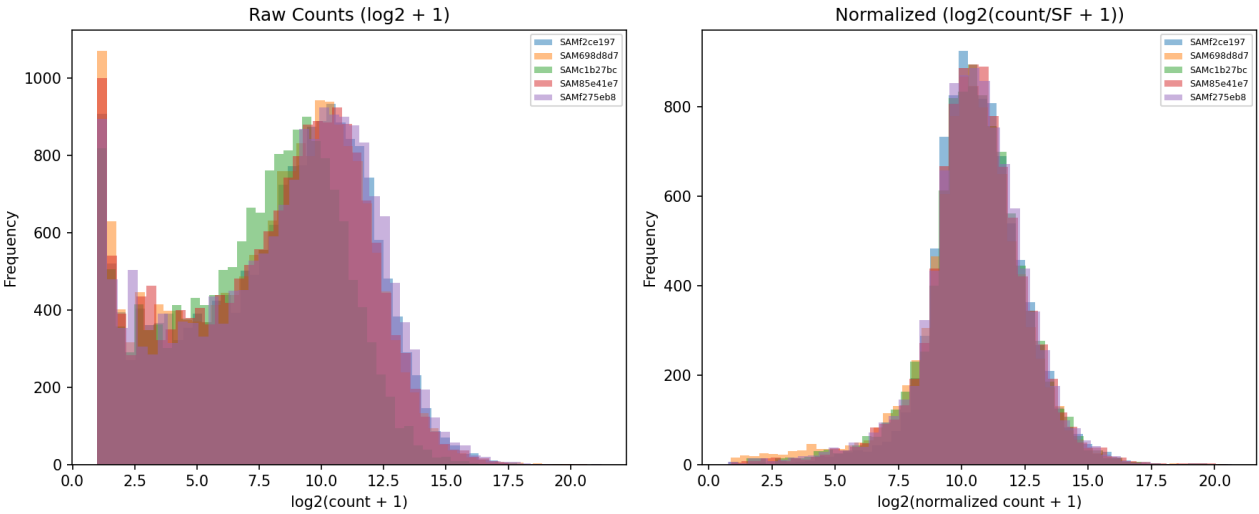


Figure: Gene expression distributions before and after normalization

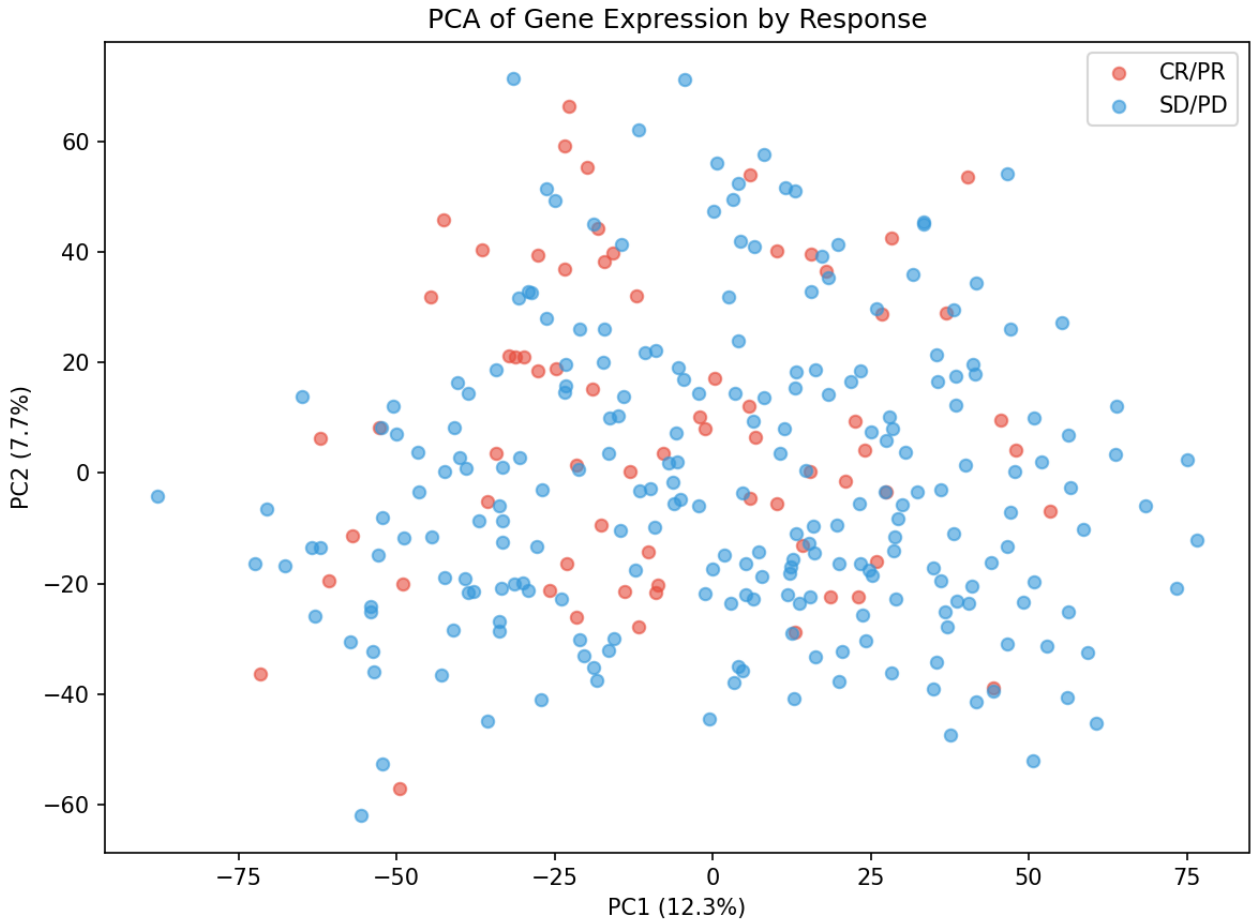


Figure: PCA of normalized gene expression coloured by response

3. Clinical Feature Associations

14 clinical variables were tested for association with response; 6 were significant at FDR < 0.05. The strongest association was Neoantigen burden per MB ($p = 5.34e-09$). Multivariate logistic regression identified 2 significant predictors ($p < 0.05$) from 10 variables. Cox proportional-hazards regression found 3 significant covariates for overall survival.

Table: Univariate Associations

variable	test	statistic	p_value	fdr
IC Level	Chi-square	11.60	0.0030	0.0085
TC Level	Chi-square	0.48	0.7860	0.8464
Immune phenotype	Chi-square	1.87	0.3920	0.4989
Sex	Fisher's exact	1.59	0.2430	0.3780
Baseline ECOG Score	Chi-square	8.79	0.0123	0.0287
TCGA Subtype	Chi-square	2.69	0.4419	0.5155
Lund	Chi-square	22.98	8.03e-04	0.0028
Met Disease Status	Chi-square	15.39	4.54e-04	0.0021
Tobacco Use History	Chi-square	3.34	0.1880	0.3290
Received platinum	Fisher's exact	0.64	0.1821	0.3290
Intravesical BCG administered	Fisher's exact	0.97	1.0000	1.0000
FMOne mutation burden per MB	Mann-Whitney U	7,680	1.14e-07	7.98e-07
Neoantigen burden per MB	Mann-Whitney U	6,626	5.34e-09	7.48e-08
sizeFactor	Mann-Whitney U	8,371	0.3779	0.4989

Table: Multivariate Logistic Regression

variable	coefficient	odds_ratio	or_ci_lower	or_ci_upper	p_value
IC Level	0.472	1.603	0.967	2.658	0.0676
immpheno_excluded	0.100	1.105	0.510	2.392	0.8003
immpheno_inflamed	-0.365	0.695	0.258	1.873	0.4714
TMB	0.120	1.128	1.075	1.183	8.48e-07
TMB_missing	-0.548	0.578	0.235	1.421	0.2323
ECOG	-0.969	0.379	0.208	0.692	0.0016
tcga_II	-0.180	0.835	0.379	1.839	0.6552
tcga_III	-0.593	0.553	0.210	1.455	0.2299
tcga_IV	0.184	1.202	0.493	2.932	0.6860
Sex (Male)	0.278	1.320	0.598	2.915	0.4920

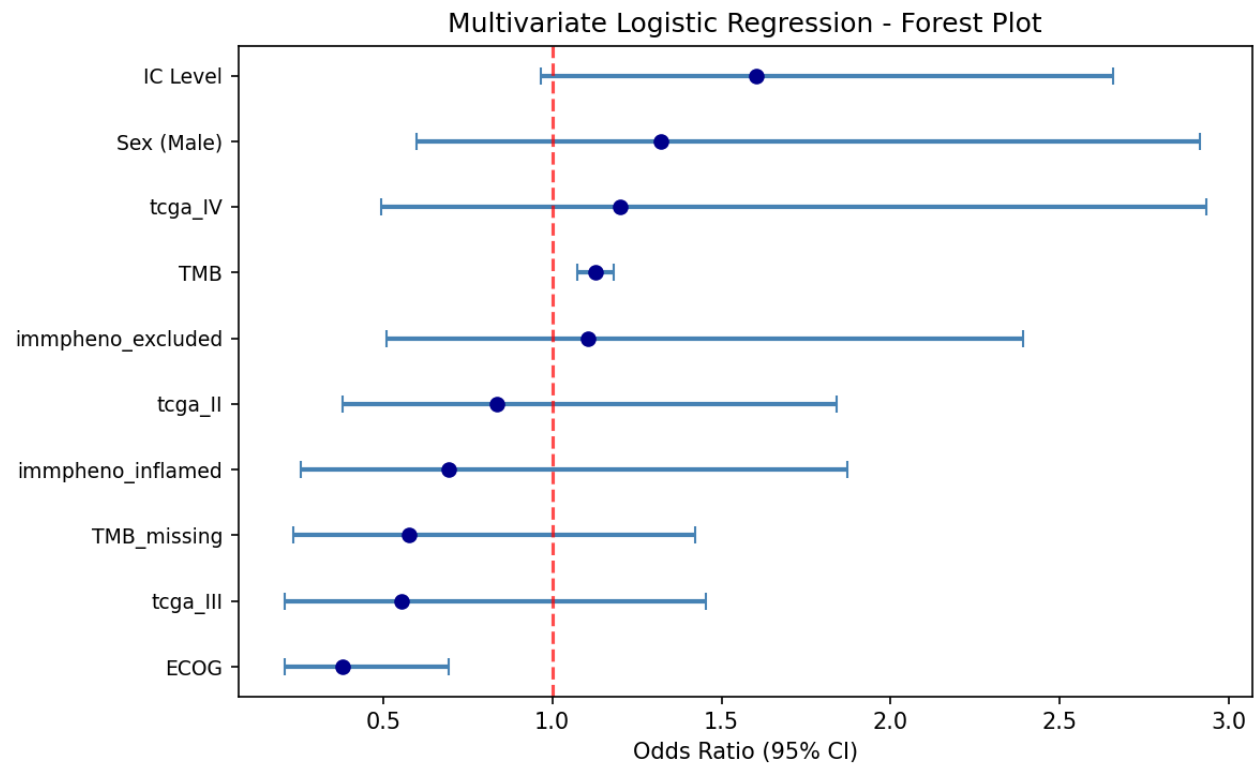


Figure: Forest plot of univariate clinical associations

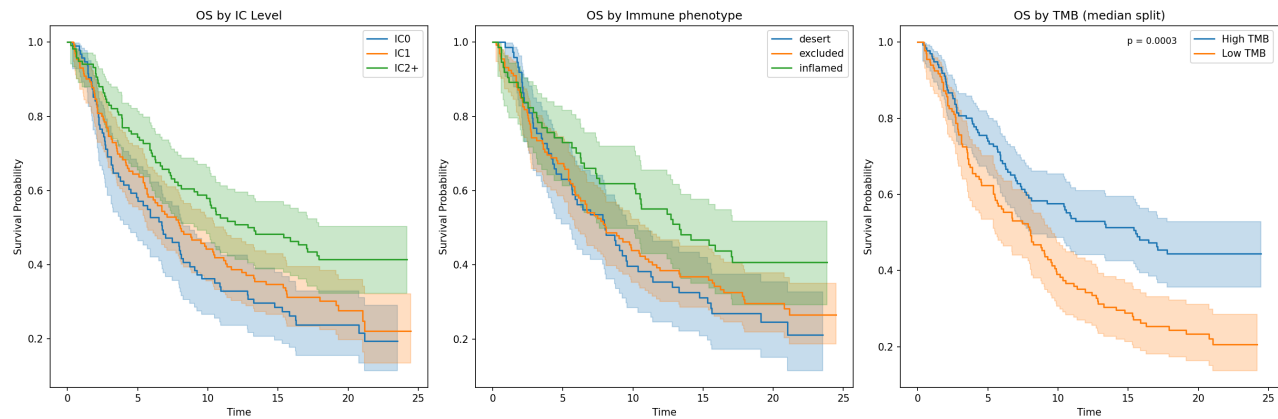


Figure: Kaplan-Meier overall survival by response

4. Genomic Alteration Landscape

28 genes were tested for mutation-response association; 0 reached $FDR < 0.05$. The top gene was FBXW7 (OR = 4.28, $p = 0.0167$). Tumour mutational burden as a response predictor: AUC=0.728.

Table: Top 15 Mutated Genes

gene	total_mutated	resp_freq	nonresp_freq	odds_ratio	p_value	fdr
FBXW7	12	0.117	0.030	4.28	0.0167	0.3464
CDKN2A	42	0.100	0.216	0.40	0.0534	0.3464
ATM	11	0.100	0.030	3.60	0.0716	0.3464
ERBB3	15	0.117	0.048	2.62	0.0755	0.3464
KDM6A	43	0.267	0.162	1.89	0.0854	0.3464
CREBBP	24	0.167	0.084	2.19	0.0877	0.3464
FRS2	18	0.133	0.060	2.42	0.0930	0.3464
ZNF703	13	0.100	0.042	2.54	0.1107	0.3464
MDM2	19	0.133	0.066	2.18	0.1114	0.3464
CDKN2B	33	0.083	0.168	0.45	0.1368	0.3830
ARID1A	54	0.300	0.216	1.56	0.2166	0.5513
EGFR	10	0.017	0.054	0.30	0.2981	0.6667
FGF10	12	0.083	0.042	2.08	0.3095	0.6667
RICTOR	16	0.100	0.060	1.74	0.3764	0.7273
STAG2	17	0.100	0.066	1.58	0.3980	0.7273

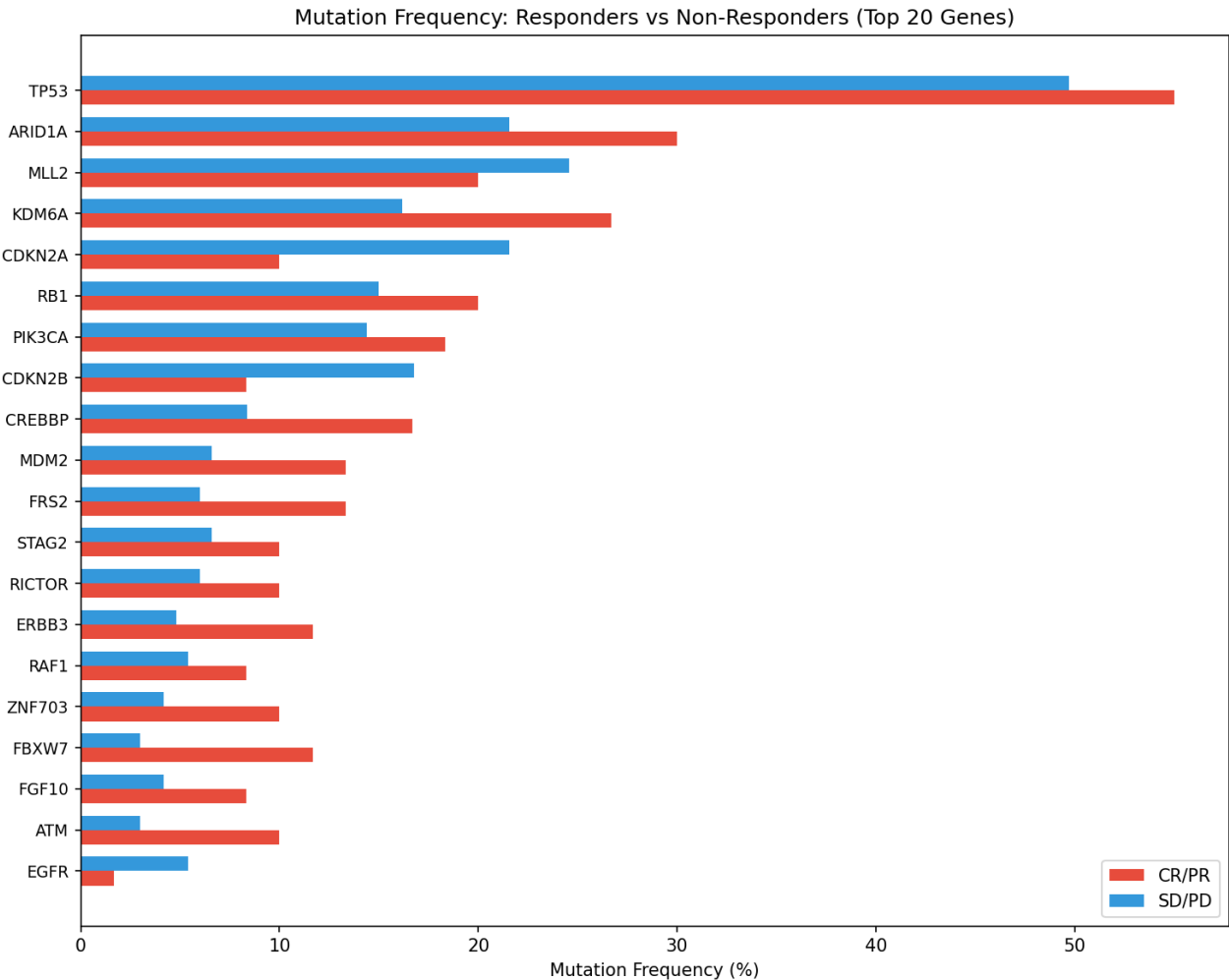


Figure: Mutation frequency in responders vs non-responders

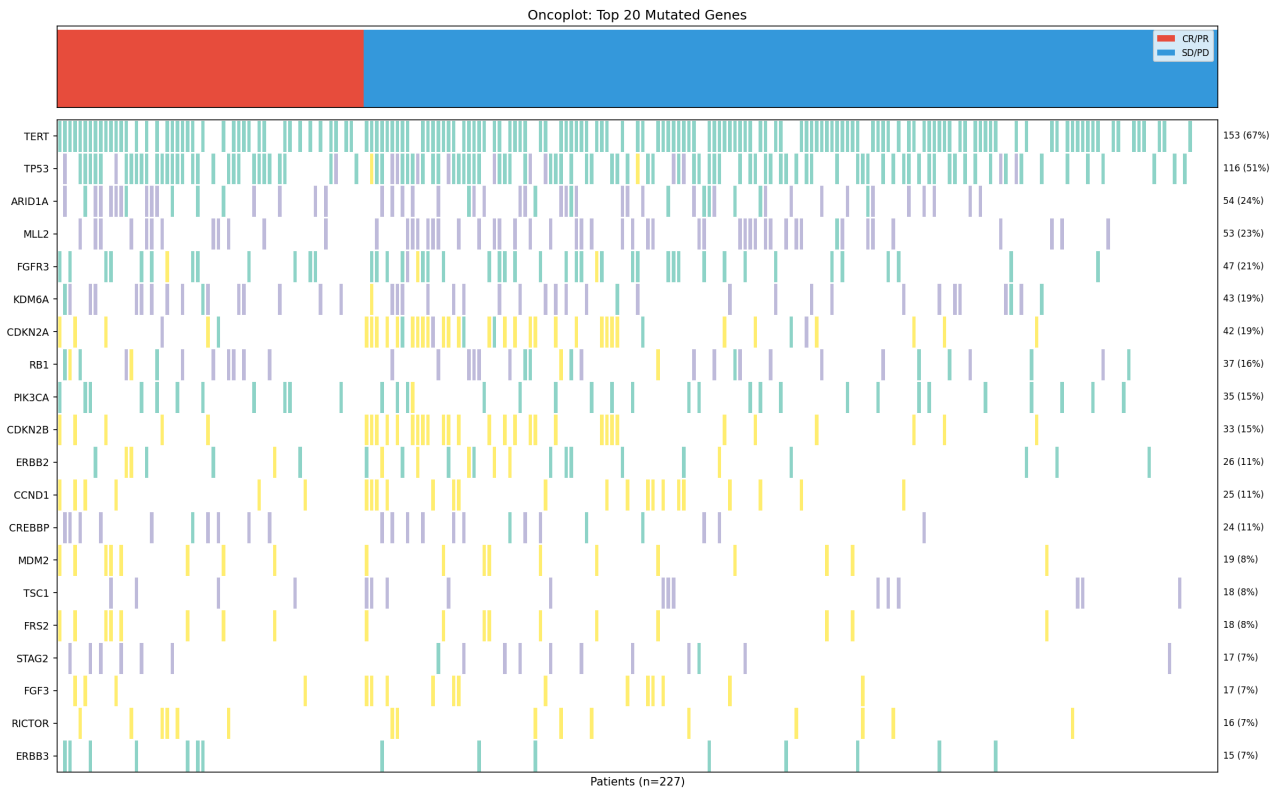


Figure: Oncoplot of top mutated genes

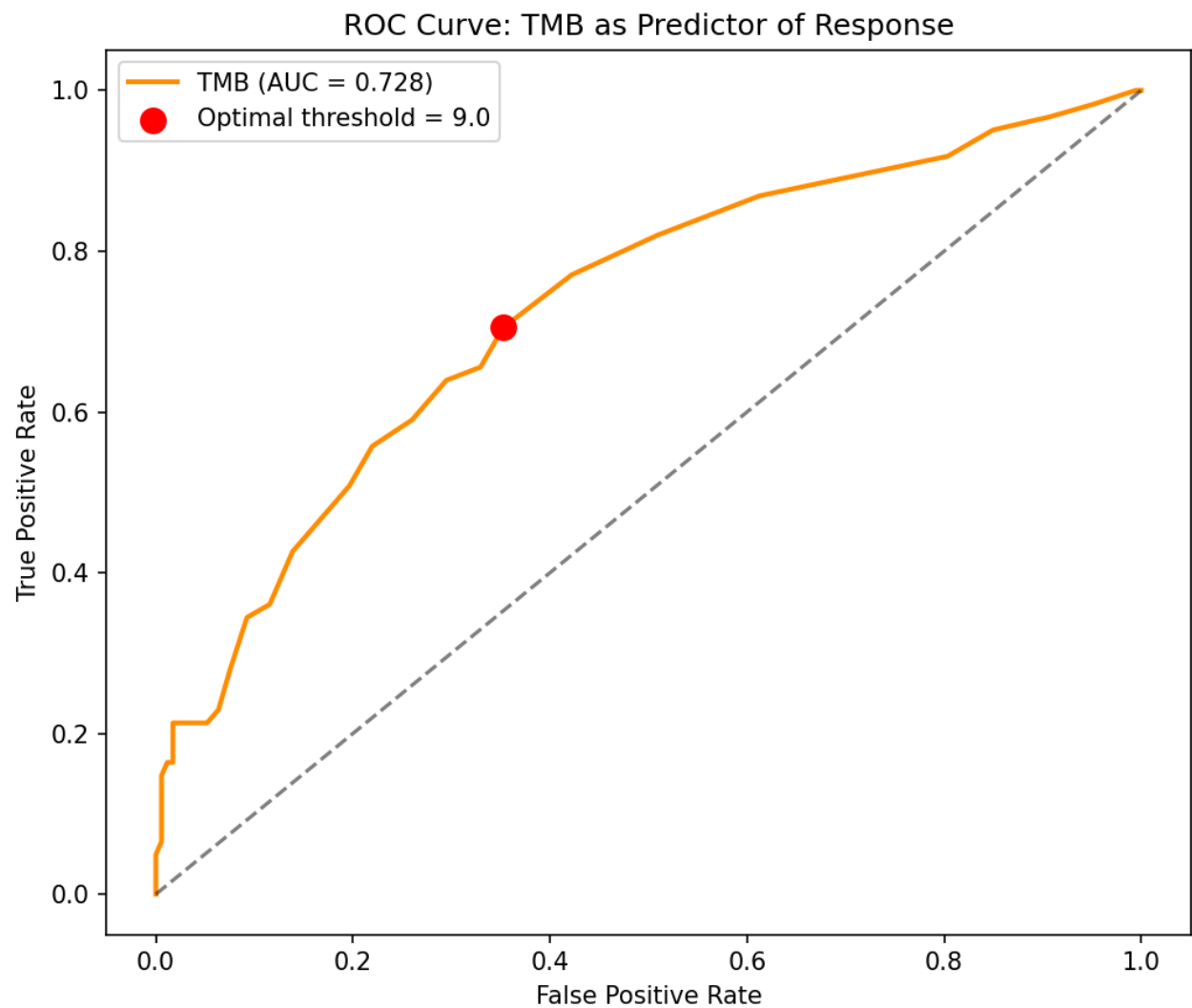


Figure: TMB as a response predictor (ROC)

5. Differential Gene Expression

Mann-Whitney U tests were performed on 7,003 genes (after independent pre-filtering) comparing responders (CR/PR) vs non-responders (SD/PD). 14 genes were significant at $|\log_2FC| > 1$ and $FDR < 0.05$ (4 up-regulated in responders, 10 down-regulated). Top up-regulated: SGOL1, PSMC4, FANCI. Top down-regulated: PCDHGC3, ZSWIM6, KCND3.

Table: Top Differentially Expressed Genes

gene	log2FC	mean_responder	mean_nonresponder	p_value	fdr
SGOL1	0.684	9.30	8.61	2.59e-06	0.0046
PSMC4	0.278	11.72	11.44	5.40e-06	0.0046
FANCI	0.455	13.22	12.76	5.48e-06	0.0046
ESPL1	0.635	10.17	9.54	5.78e-06	0.0046
POLA2	0.382	10.25	9.87	5.87e-06	0.0046
DSN1	0.376	9.77	9.39	5.96e-06	0.0046
RFC5	0.422	9.95	9.52	1.09e-05	0.0070
NUF2	0.737	9.92	9.19	1.29e-05	0.0075
GPATCH1	0.253	10.21	9.95	1.83e-05	0.0076
SPAG5	0.530	11.74	11.21	1.91e-05	0.0076
PCDHGC3	-0.597	9.34	9.94	2.70e-06	0.0046
ZSWIM6	-0.352	10.09	10.44	3.33e-06	0.0046
KCND3	-1.060	6.97	8.03	5.32e-06	0.0046
CHST3	-0.645	7.87	8.51	1.00e-05	0.0070
ACVR1	-0.314	10.90	11.22	2.79e-05	0.0076
SH3TC1	-0.624	8.48	9.11	3.50e-05	0.0076
MMP10	-1.643	6.06	7.70	3.61e-05	0.0076
EXT1	-0.408	11.79	12.20	3.88e-05	0.0078
RAPGEF3	-0.676	7.91	8.58	4.25e-05	0.0080
TGFB1	-0.478	10.40	10.88	4.77e-05	0.0080

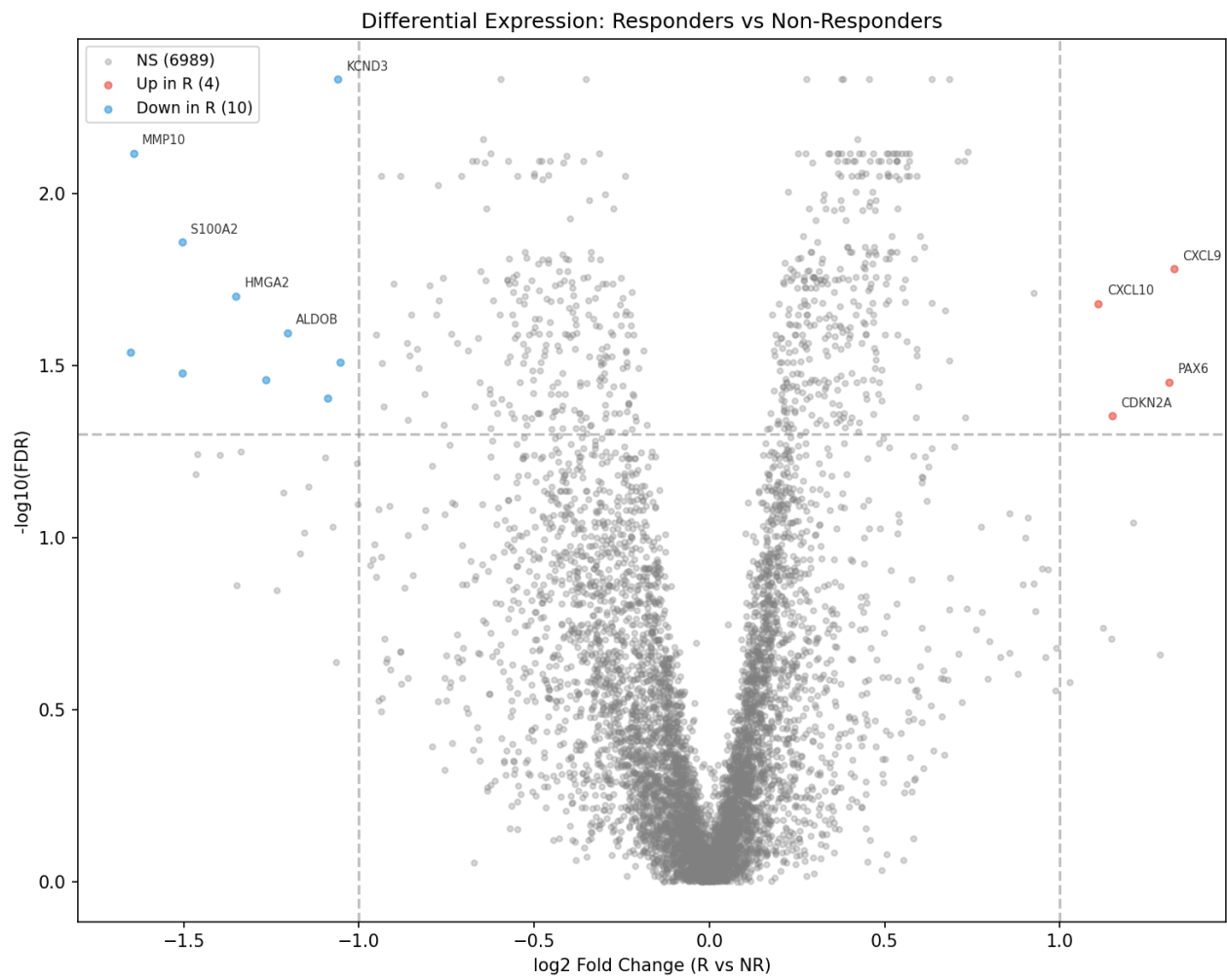


Figure: Volcano plot of differential expression

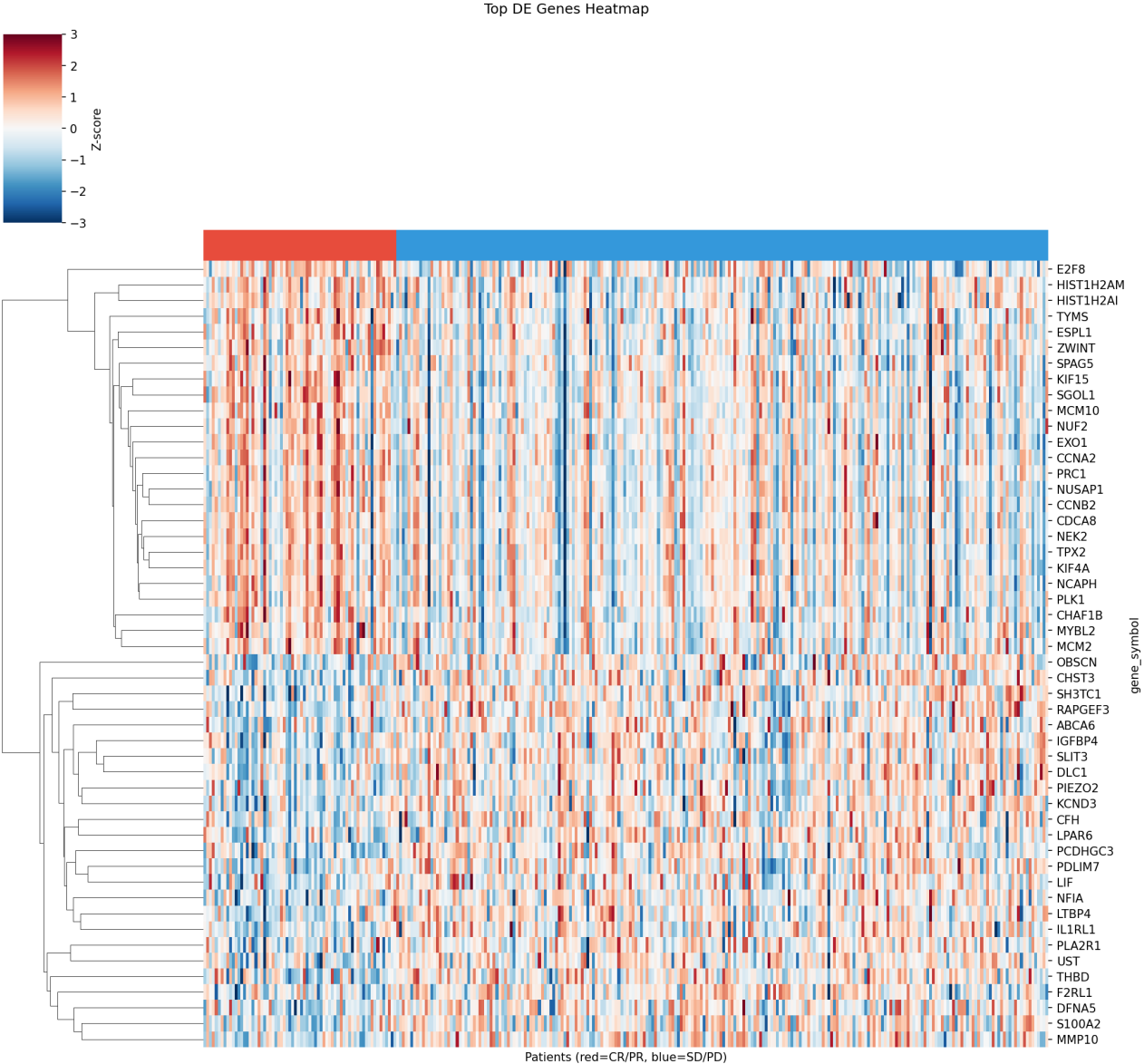


Figure: Heatmap of top differentially expressed genes

6. Pathway Enrichment Analysis

Over-Representation Analysis (ORA, Fisher's exact test) was run across 4 gene set collection(s). 421 gene sets were enriched at $FDR < 0.05$. Top ORA hits: HALLMARK_E2F_TARGETS (Upregulated in R), HALLMARK_G2M_CHECKPOINT (Upregulated in R), HALLMARK_MITOTIC_SPINDLE (Upregulated in R). Pre-ranked GSEA (1,000 permutations) was run across 4 collection(s). 1717 gene sets reached $FDR < 0.05$. Top GSEA hits: HALLMARK_E2F_TARGETS (NES=3.15), REACTOME_CELL_CYCLE (NES=3.12), REACTOME_CELL_CYCLE_CHECKPOINTS (NES=3.11).

Table: Top ORA Enrichments

collection	direction	term	overlap	term_size	p_value	fdr
Hallmark	Upregulated in R	HALLMARK_E2F_TARGETS	50	134	2.91e-44	2.18e-42
Hallmark	Upregulated in R	HALLMARK_G2M_CHECKPOINT	43	118	2.00e-37	7.51e-36
Hallmark	Upregulated in R	HALLMARK_MITOTIC_SPINDLE	27	134	3.73e-16	9.32e-15
Hallmark	Upregulated in R	HALLMARK_SPERMATOGENESIS	13	42	7.02e-11	1.32e-09
Hallmark	Upregulated in R	HALLMARK_MYC_TARGETS_V1	16	124	3.87e-07	4.84e-06
Hallmark	Upregulated in R	HALLMARK_DNA_REPAIR	13	81	3.88e-07	4.84e-06
Hallmark	Downregulated in R	HALLMARK_TGF_BETA_SIGNALING	7	40	1.17e-04	0.0013
Hallmark	Downregulated in R	HALLMARK_COAGULATION	9	83	5.57e-04	0.0052
Hallmark	Downregulated in R	HALLMARK_TNFA_SIGNALING_VIA_NF	12	144	8.05e-04	0.0067
Hallmark	Downregulated in R	HALLMARK_P53_PATHWAY	10	135	0.0051	0.0384

Table: Top GSEA Enrichments

collection	term	es	nes	size	p_value	fdr
Hallmark	HALLMARK_E2F_TARGETS	0.814	3.152	134	9.99e-04	0.0023
Reactome	REACTOME_CELL_CYCLE	0.727	3.125	324	9.99e-04	0.0031
Reactome	REACTOME_CELL_CYCLE_CHECKPOINTS	0.795	3.108	141	9.99e-04	0.0031
GO_BP	GOBP_CHROMOSOME_SEGREGATION	0.765	3.058	179	9.99e-04	0.0037
Reactome	REACTOME_CELL_CYCLE_MITOTIC	0.724	3.046	270	9.99e-04	0.0031
GO_BP	GOBP_CHROMOSOME_ORGANIZATION	0.718	3.027	268	9.99e-04	0.0037
GO_BP	GOBP_NUCLEAR_CHROMOSOME_SEGREGATION	0.777	3.017	138	9.99e-04	0.0037
Hallmark	HALLMARK_G2M_CHECKPOINT	0.773	2.938	118	9.99e-04	0.0023
GO_BP	GOBP_DNA_TEMPLATED_DNA_REPLICATION	0.819	2.933	88	9.99e-04	0.0037
GO_BP	GOBP_DNA_REPAIR	0.701	2.932	242	9.99e-04	0.0037

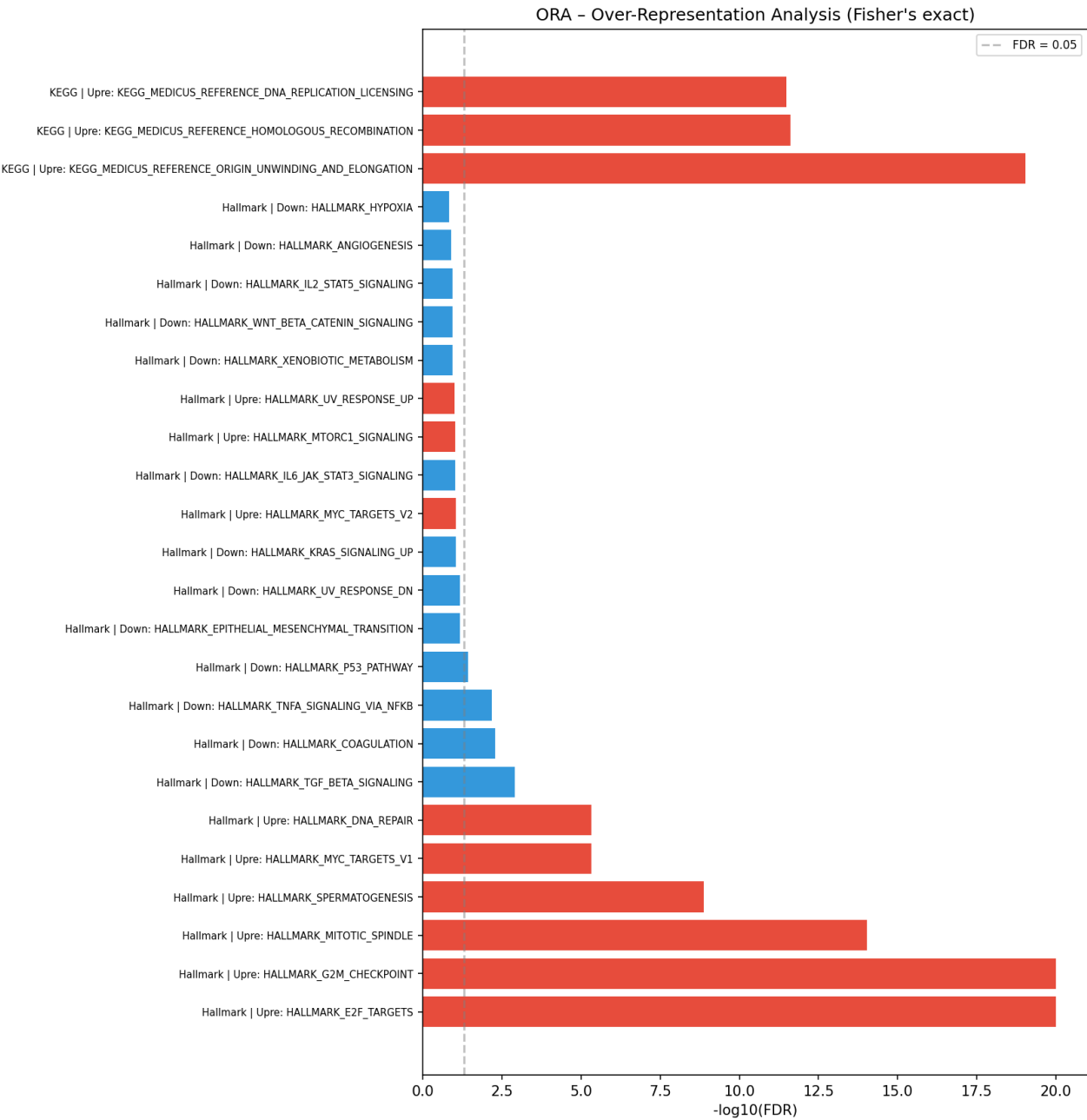


Figure: ORA enrichment bar plot

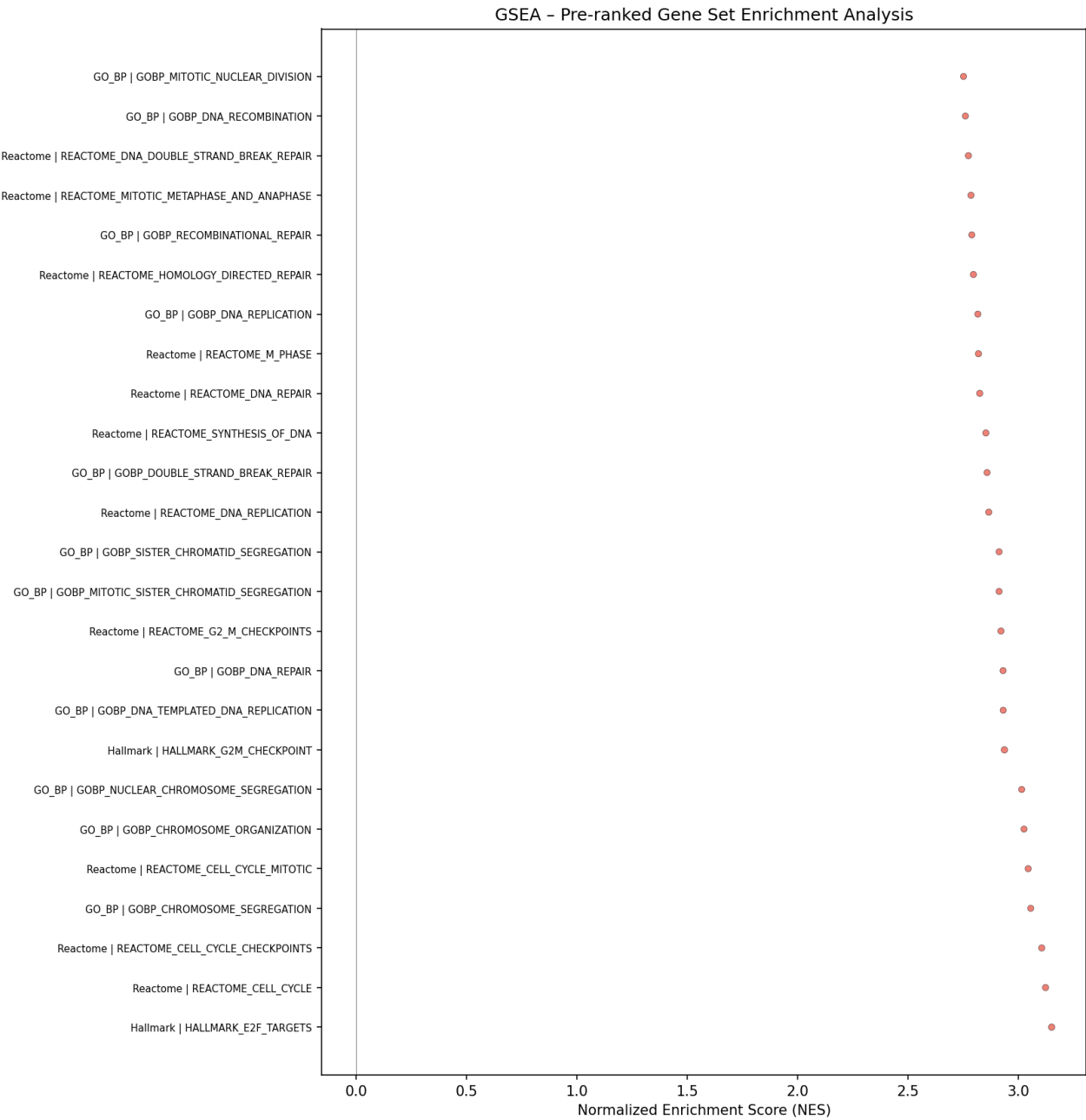


Figure: GSEA enrichment dot plot

7. Predictive Gene Expression Signature

Elastic Net regularised logistic regression selected 96 signature genes via nested cross-validation. Top positive predictors (higher in R): IFT74, TMED3, AKR1B1. Top negative predictors (higher in NR): CHST3, DHRS1, AP3M2. Random Forest identified 30 important features. 7 genes overlap between Elastic Net and Random Forest selections. Model comparison (5-fold nested CV): Elastic Net: AUC = 0.714 ± 0.055 Random Forest: AUC = 0.701 ± 0.056 TMB: AUC = 0.728 ± 0.000 PD-L1 IC: AUC = 0.618 ± 0.000

Table: Model Comparison

model	mean_auc	std_auc
Elastic Net	0.714	0.055
Random Forest	0.701	0.056
TMB	0.728	0.000
PD-L1 IC	0.618	0.000

Table: Top 15 Elastic Net Signature Genes

gene	coefficient	rank
IFT74	0.2779	1
CHST3	-0.2692	2
DHRS1	-0.2506	3
AP3M2	-0.2321	4
AIM1	-0.2282	5
GLYCTK	-0.2224	6
TMED3	0.2125	7
AKR1B1	0.2055	8
EML1	0.2053	9
YAP1	-0.2045	10
TOM1L1	0.1991	11
MERTK	-0.1972	12
CXCL10	0.1904	13
ITGAE	0.1577	14
ZNF649	-0.1538	15

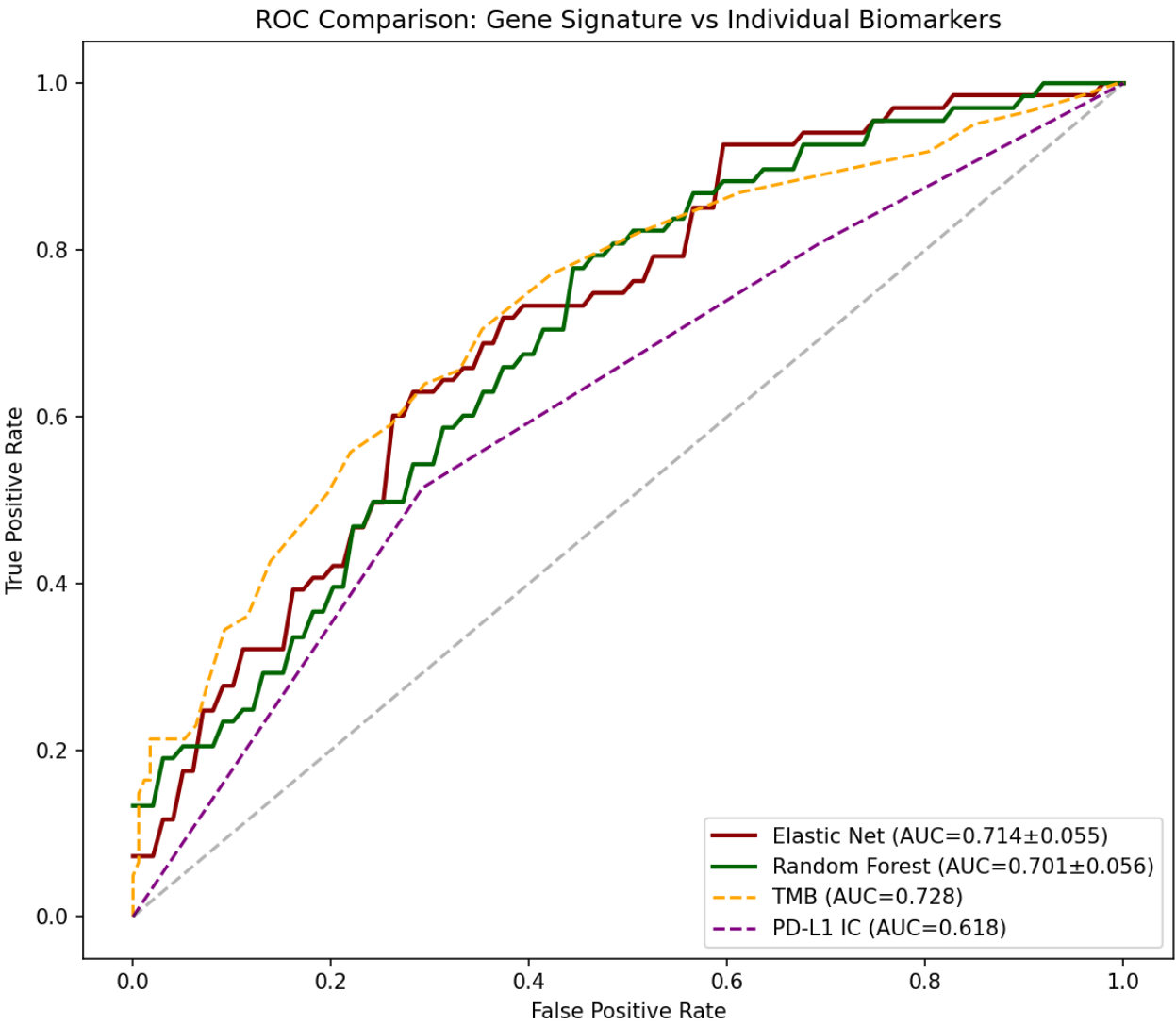


Figure: ROC comparison across predictive models

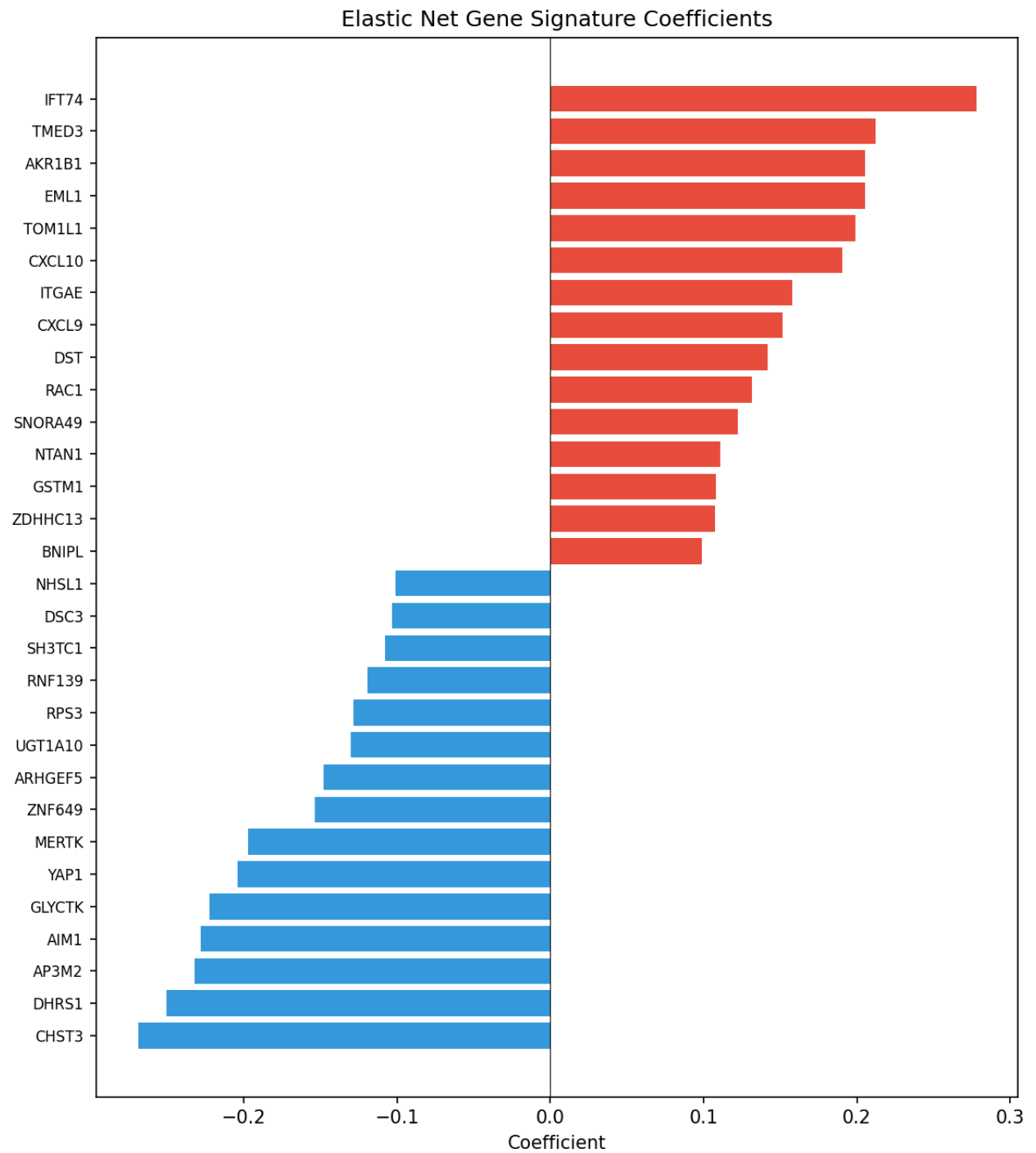


Figure: Elastic Net signature gene coefficients

8. Integrated Model & Survival Analysis

The best-performing classifier was Integrated LR with a mean AUC of 0.990. Expression-only: AUC = 0.988, Clinical-only: AUC = 0.721. Survival model C-indices: Expression-only: C-index = 0.6088 Clinical-only: C-index = 0.6576 Integrated: C-index = 0.6684

Table: Classifier Performance

model	mean_auc	std_auc	mean_accuracy	mean_f1
Expression-only LR	0.988	0.013	0.946	0.876
Clinical-only LR	0.721	0.086	0.798	0.357
Integrated LR	0.990	0.010	0.960	0.907
Integrated GB	0.980	0.022	0.946	0.879

Table: Survival Model C-Indices

model	c_index
Expression-only	0.6088
Clinical-only	0.6576
Integrated	0.6684

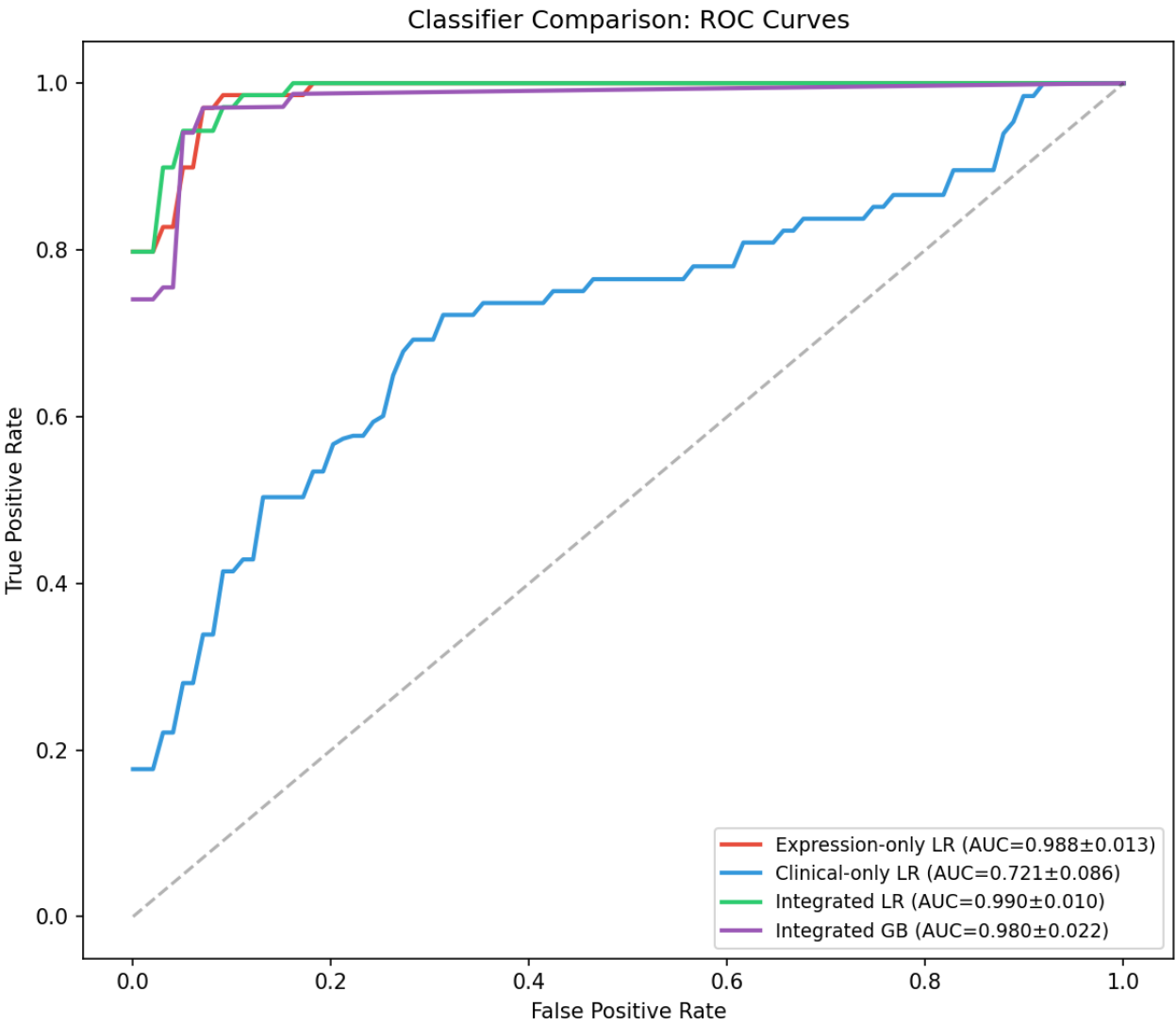


Figure: Integrated classifier ROC curves



Figure: Kaplan-Meier survival by integrated risk score

9. Methods Summary

Data source: IMvigor210 Phase-II clinical trial of atezolizumab (anti-PD-L1) in metastatic urothelial carcinoma. Gene expression was normalised using DESeq2-style size factors ($\log_2(\text{count}/\text{sizeFactor} + 1)$) and low-expression genes were filtered (minimum 10 counts in at least 15 samples). Prior to differential expression testing, an independent pre-filter removed genes with mean $\log_2$ expression < 1.0 and genes in the bottom 25% of expression variance, reducing the multiple-testing burden while preserving FDR control.

Differential expression was assessed using Mann-Whitney U tests (two-sided) comparing responders (CR/PR) vs non-responders (SD/PD). P-values were corrected for multiple testing using the Benjamini-Hochberg FDR procedure.

Over-Representation Analysis (ORA) used Fisher's exact test (one-sided, greater) to test whether up- or down-regulated gene lists were enriched in curated gene sets from MSigDB v2024.1 (Hallmark, KEGG Medicus, Reactome, GO Biological Process). Gene sets were filtered to 15-500 genes overlapping the tested gene universe. FDR correction was applied within each analysis.

Pre-ranked Gene Set Enrichment Analysis (GSEA) used the ranking metric $\text{sign}(\log_2\text{FC}) \times -\log_{10}(\text{p-value})$. Enrichment scores were computed via a weighted running-sum statistic, normalised against 1,000 permutation-based null distributions (cached per gene-set size for efficiency). Empirical p-values and FDR were calculated from the permutation distributions.

Predictive gene expression signatures were built using Elastic Net regularised logistic regression and Random Forest classifiers within a nested 5-fold cross-validation framework. Model performance was compared by mean AUC across outer folds.

Integrated classifiers combined expression-derived signature scores, clinical features, and genomic features (TMB, key mutations) using logistic regression and gradient boosting. Survival analysis used Cox proportional-hazards models, with concordance index (C-index) as the performance metric.